MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024
First Semester

Discipline Specific Core Course -MG1DSCECC101, MG1DSCCSC100 - ART OF COMPUTING AND PROBLEM SOLVING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Scheme for Evaluation

Part A: Very Short Answer Questions (Answer all) - (10 * 1 = 10 Marks)

1. **How many bytes are there in a megabyte (MB)?** [U] [1]

   Answer: 1 MB = 1024 KB = 1024 * 1024 bytes (1 mark for the correct answer)

2. **Identify the characteristics of a computer that allow it to process large volumes of data quickly.** [U] [1]

   Answer: Any two characteristics (0.5 marks each):

   - High processing speed
   - Large storage capacity
   - Ability to execute complex calculations
   - Accuracy and reliability

3. **Name the error that you encounter if you try to access an array element out of its bounds.** [K] [1]

   Answer: Runtime Errors or similar concept (1 mark)

4. **Name the type of software that is designed to manage and control the hardware components of a computer.** [K] [1]

   Answer: Operating System / System Sofware (1 mark)

5. **Which symbol indicates that the line is a preprocessor directive?** [K] [2]

Answer: # (1 mark)

6. **Identify the use of '&&' operator.** [U] [2]

   Answer: Logical AND operator (1 mark)

7. **Illustrate how the syntax of a for loop is structured?** [U] [3]

   Answer: for (initialization; condition; increment/decrement) (1 mark)

8. **Describe the outcome of the following expression: 5+2*3** [U] [2]

   Answer: 11 (1 mark)

9. **How do you declare a string in C?** [K] [3]

   Answer: char string_name[size]; (1 mark)

10. **How do you skip to the next iteration of a loop?** [U] [3]

    Answer: continue; (1 mark)

Part B: Short Answer Questions (4 out of 6 Questions) - (4 * 5 = 20 Marks)

11. **List the steps of the problem-solving life cycle and briefly explain each step.** [K] [1]

    Answer: (1 mark for listing the steps, 4 marks for explanations)

    - Problem Definition
    - Problem Analysis
    - Algorithm Design
    - Coding
    - Testing and Debugging
    - Documentation

12. **Write an algorithm to find the sum of the first n natural numbers.** [U] [1]

    Answer: (5 marks for a correct algorithm with initialization, loop, and calculation)

    Step 1: Start
    Step 2: Input n
    Step 3: sum = 0, i = 1
    Step 4: Repeat steps 4.1 to 4.2 while i <= n
    4.1: sum = sum + i
    4.2: i = i + 1

Step 5: Print sum
Step 6: Stop

13. **Demonstrate the use of at least five keywords in C with an application program.** [U] [2]
    Answer: (1 mark for each keyword with correct usage in a C program)
    - int, float, char, if, for, while, return, etc.
14. **Differentiate goto statement and structured control flow statements in C.** [U] [3]

    Answer: (2.5 marks for explaining each concept with differences)

    - goto: Unconditional jump, can lead to spaghetti code.
    - Structured control flow (if, else, for, while): More organized and readable code.
15. **Discuss the commonly used string handling functions in C, highlighting their purpose and usage with examples.** [U] [3]

    Answer: (1 mark for each function with explanation and example)

    - strlen(), strcpy(), strcat(), strcmp(), etc.
16. **When would you use the break statement in loops? Explain with an example.** [U] [3]

    Answer: (2 marks for explanation, 3 marks for a relevant example)

    - To exit a loop prematurely based on a condition.

Part C: Essay Questions (Answer any 2 out of 3 Questions) - (2 * 10 = 20 Marks)

17. **Explain the differences between syntax errors, logical errors, and runtime errors. Provide examples for each type of error and how they can be resolved.** [U] [1]

    Answer: (2 marks for each type of error with explanation and example, and 1 mark for explaining resolution methods)

    - **Syntax errors:** Violations of the programming language rules.
    - **Logical errors:** Errors in the program's logic or algorithm.
    - **Runtime errors:** Errors that occur during program execution.
18. **Discuss the use of variables and constants in programming. Explain different types of constants with examples.** [K] [2]

Answer: (4 marks for explaining variables and their use, 1.5 marks for each type of constant with explanation and example)

- **Variables:** Symbolic names for memory locations that store data.
- **Constants:** Values that do not change during program execution.
  - Integer constants, floating-point constants, character constants, string literals, etc.

19. **Explain the different forms of 'if' statements with syntax and examples.** [U] [3]

Answer: (2.5 marks for each form of if statement with syntax and example)

- if statement
- if-else statement
- if-else if-else statement
- Nested if statements